Arithmetic Average, Median, Mode and Expected value (Mean)

Definition: Arithmetic Average and Median

- Definition: The arithmetic average of n numbers is the sum of n numbers divided by n

$$\text{Arithmetic average} = \frac{\sum_{i=1}^{n} x_i}{n}$$

  - It can be used with discrete data.
  - It has an influence of outlier number in your set.

Value	15	18	16	14	15	15	12	17	**90**	**95**

  - Average Value is 30.7 with 90 and 95. Average Value is 15.25 without 90 and 95

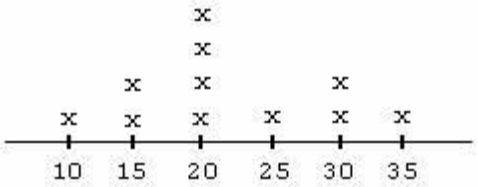
- Definition: The median is a number in the middle of data set
  - To find the median, your numbers have to be listed in numerical order
  - The median is less affected by outliers
  - Median is pretty close to the value you expect

Value	15	18	16	14	15	15	12	17	**90**	**95**
Ordered Value	12	14	15	15	15	16	17	18	90	95

Definition: Mode

- Definition: The mode of a set of numbers is the most common number

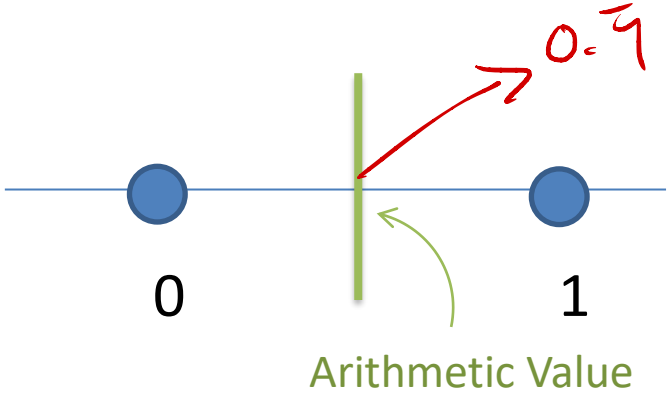
10, 20, 15, 20, 25, 30, 35, 20, 20, 30, 15



Method	Adv.	Disadv.
Arithmetic averaging	Takes account of all values to calculate the average	Very minute or large values can affect the average
Median	Not affected by very large or very small values	Arranging the data is time consuming in the case of large number of observation
Mode	The only averages that can be used if the data set is not in numbers	There can be more than one mode

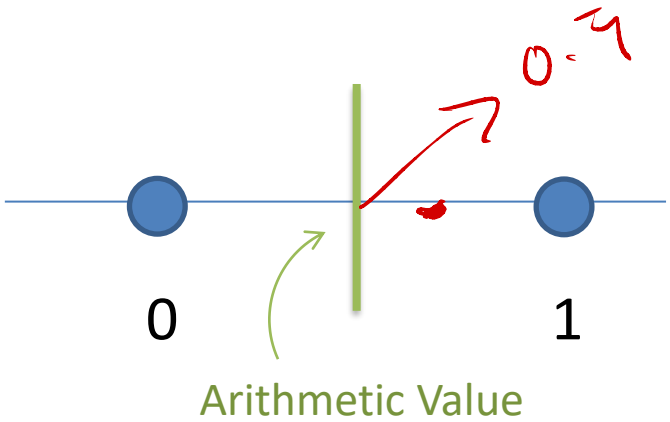
If a random variable X has

$$P_X(x) = \begin{cases} 1/2 & x = 0 \\ 1/2 & x = 1 \\ 0 & otherwise \end{cases}$$



If a random variable Y has

$$P_Y(y) = \begin{cases} 1/3 & y = 0 \\ 2/3 & y = 1 \\ 0 & otherwise \end{cases}$$



Definition: Expected Value (Expectation) or Mean

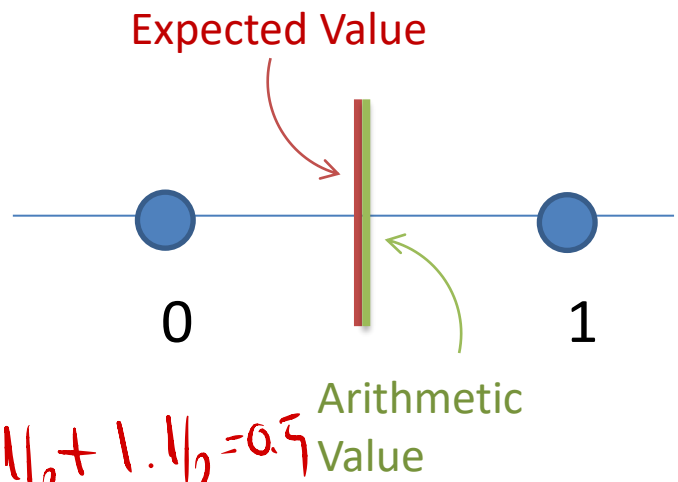
Definition: The expected value of random variable X

$$E(X) = \mu_X = \sum_{x \in S_X} x.P_X(x)$$

$$E(X) = \mu_X = \int_{all\ x} x.f_X(x)$$(will be discussed in Chapter 3)

If a random variable X has

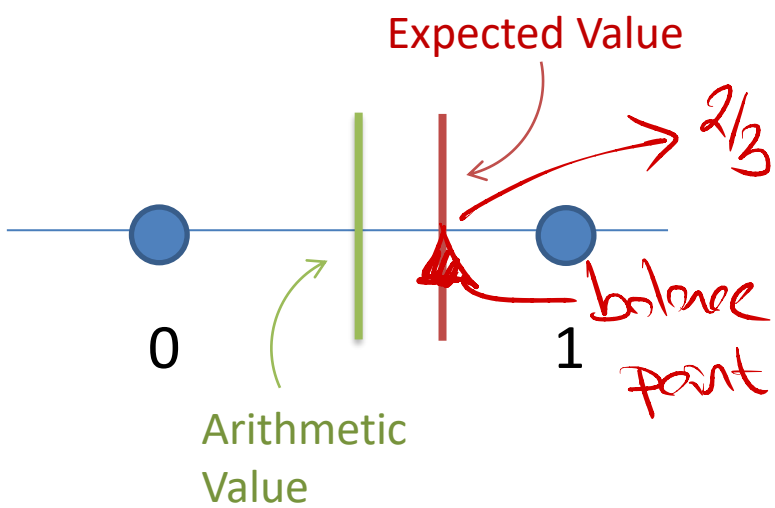
$$P_X(x) = \begin{cases} 1/2 & x = 0 \\ 1/2 & x = 1 \\ 0 & otherwise \end{cases}$$



$$E(X) = \mu_X = \sum_{x=0,1} x \cdot P_X(x) \Rightarrow 0 \cdot 1/2 + 1 \cdot 1/2 = 0.5$$

If a random variable Y has

$$P_Y(y) = \begin{cases} 1/3 & y = 0 \\ 2/3 & y = 1 \\ 0 & otherwise \end{cases}$$



$$E(Y) = \mu_Y = 0 \cdot 1/3 + 1 \cdot 2/3$$

Instructive example

Find the expected value of the probability distribution of the number of heads obtained in 3 flips. Note that: the coin lands head with probability ½

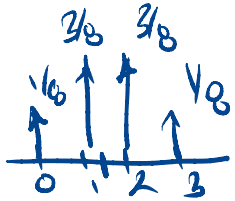
$X = \#$ of heads in 3 flips $\quad S_X = \{0, 1, 2, 3\}$

$X=0 \Rightarrow \frac{1}{2}\ \frac{1}{2}\ \frac{1}{2} \Rightarrow 1/8 \times 1$

$X=1 \Rightarrow \frac{1}{2}\ \frac{1}{2}\ \frac{1}{2} \Rightarrow 1/8 \times 3$

$X=2 \Rightarrow \frac{1}{2}\ \frac{1}{2}\ \frac{1}{2} \Rightarrow 1/8 \times 3$

$X=3 \Rightarrow \frac{1}{2}\ \frac{1}{2}\ \frac{1}{2} \Rightarrow 1/8 \times 1$



$E(X) = \mu_X = 0 \cdot 1/8 + 1 \cdot 3/8 + 2 \cdot 3/8 + 3 \cdot 1/8 = 3/2 = 1.5$

Find the expected value of the probability distribution of the number of heads obtained in 3 flips. Note that: the coin lands head with probability 2/3

$X=0 \quad 1/3\ 1/3\ 1/3 \cdot 1$

$X=1 \quad 2/3\ 1/3\ 1/3 \cdot 3$

$X=2 \quad 2/3\ 2/3\ 1/3 \cdot 3$

$X=3 \quad 2/3\ 2/3\ 2/3 \cdot 1$

$E(X) = 0 \cdot 1/27 + 1 \cdot 6/27 + 2 \cdot 12/27 + 3 \cdot 8/27 = 2$

Theorem 2.3: expected value for common distributions

The bernoulli random variable X has expected value $E[X] = p$

The geometric random variable X has expected value $E[X] = 1/p$

The binomial random variable X has expected value $E[X] = np$

The poisson random variable X has expected value $\mathrm{E}[X] = \lambda$

The uniform random variable X has expected value $E[X] = (min. + max.)/2$

Example-2.9

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by inspector. Find the expected value of the number of good components in this sample.

$X = \#$ of good components $\quad S_X = \{0, 1, 2, 3\}$

$X = 0 \quad \frac{3}{7} \quad \frac{2}{6} \quad \frac{1}{5} \quad (1)$

$X = 1 \quad \frac{4}{7} \quad \frac{3}{6} \quad \frac{2}{5} \quad (3)$

$X = 2 \quad \frac{4}{7} \quad \frac{3}{6} \quad \frac{3}{5} \quad (3)$

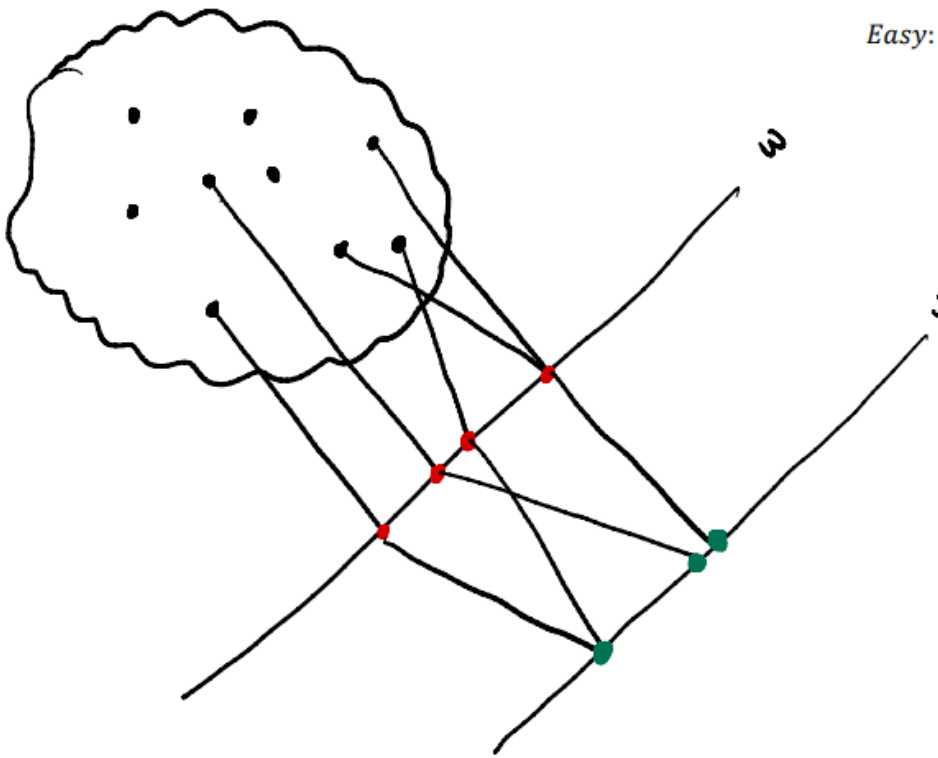
$X = 3 \quad \frac{4}{7} \quad \frac{3}{6} \quad \frac{2}{5} \quad (1)$

$$E(X) = 0 \cdot \frac{6}{210} + 1 \cdot \frac{72}{210} + 2 \cdot \frac{108}{210} + 3 \cdot \frac{24}{210}$$

Expected value of a Function

$$Hard:\ E[Y] = \mu_Y = \sum_{y \in S_Y} y P_Y(y)$$

$$Easy:\ E[Y] = \mu_Y = \sum_{w \in S_w} g(w).P_W(w)$$



$y = g(w)$ → transfer function

$= \underbrace{2w+1}_{\text{linear}}$ or $\underbrace{w^2}_{\text{non-linear}}$

for $w = 2$ $\quad y = g(2)$

Theorem 2.4:

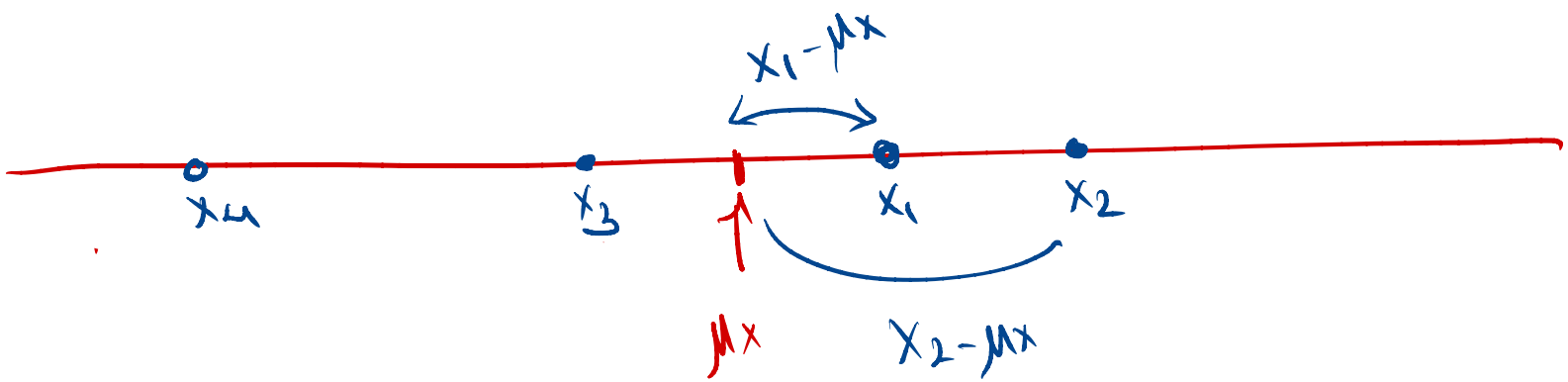
In general

$$E(g(x)) \neq g\big(E(x)\big)$$

Theorem 2.5:

For any random variable X (discr. or cont.)

$$E[X - \mu_X] = 0$$



$$E[Y] = E[g(x)] = \sum_{x \in S_X} g(x) P_X(x)$$

Theorem 2.6: Linearity of Expectation

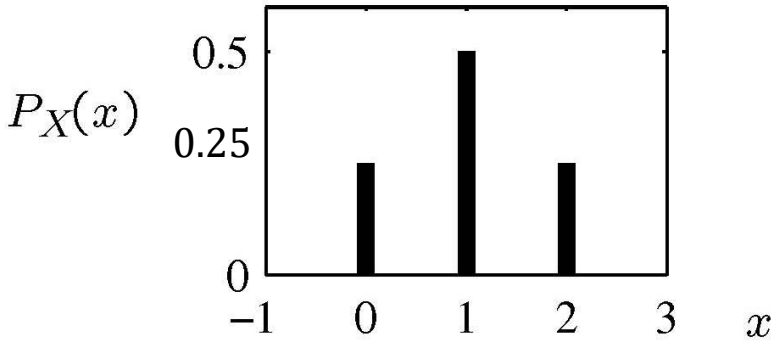
For any random variable X (discr. or cont.)

$$E[aX \mp b] = aE[X] \mp b$$

$$E(aX+b) \Rightarrow E(aX)+E(b)$$

$$aE(X)+b$$

Example-2.10



What is the expected value of $V = g(X) = 4X + 7$?

$$E(V) = E(4X+7)$$
$$= 4E(X) + 7$$
$$= 4 + 7 = \underline{11}$$

let $W = h(X) = X^2$. What is E[W]?

$$E(W) = E(X^2) = \sum_{X^2:\,0,1,4} x^2 P_X(x) = 0 \cdot 1/4 + 1 \cdot 1/2 + 4 \cdot 1/4$$

You draw one card from a standard deck of playing cards. If you pick a heart, you will win \$10. If you pick a face card, which is not a heart, you win \$8. If you pick any other card, you lose \$6. Do you want to play? Explain

$X =$ money you win $\quad S_X = \{10, 8, -6\}$

$P(X=10) = 13/52 \quad P(X=8) = 9/52 \quad P(X=-6) = 30/52$

$E(X) = 10 \cdot 13/52 + 8 \cdot 9/52 + (-6)\, 30/52 = 0.42\$$

You pay \$10 to play the following game of chance. There is a bag containing 12 balls, five are red, three are green and the rest are yellow. You are to draw one ball from the bag. You will win \$14 if you draw a red ball and you will win \$12 is you draw a yellow ball. How much do you expect to win or loss if you play this game 100 times?

$X =$ net winnings $\quad S_X = \{4, 2, -10\}$

$P(X=4) = 5/12 \quad P(X=2) = 4/12 \quad P(X=-10) = 3/12$

$E(X) = 4 \cdot 5/12 + 2 \cdot 4/12 + (-10) \cdot 3/12 = -0.1667\$$

$100 \times (-0.1667) = -16.67\$$